PROJECT REPORT

Library Management System

Python Console Application

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Course : Python Full Stack Development

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1. Abstract

The Library Management System is a Python console-based application developed using modular programming, Object-Oriented Programming (OOP), functions, and file handling. The system automates book management, member registration, issue and return operations, and report generation. It minimizes manual work, improves record accuracy, and provides an organized way to manage library resources.

2. Introduction

Libraries require efficient management of books, members, and transactions. Manual record keeping is time-consuming and prone to errors. This project provides a simple digital solution for managing library operations using Python.

3. Problem Statement

Maintaining library records manually often results in misplaced books, inaccurate issue records, and delays in searching information. A computerized system improves efficiency and reduces human errors.

4. Objectives

- Develop a modular Library Management System using Python.
- Manage books and members efficiently.
- Issue and return books.
- Maintain records using file handling.
- Generate library reports.

5. Scope of the Project

The system is suitable for schools, colleges, training institutes, and small public libraries. It can be extended into a web application using Flask or Django.

6. Technologies Used

Technology	Purpose
Python	Core Programming
OOP	Application Design
Functions	Business Logic
File Handling	Store Book & Member Records
VS Code	Development Environment
Git & GitHub	Version Control

7. Modules

Admin

- Login
- Manage Books
- Manage Members
- Generate Reports

Book Management

- Add Book
- View Books
- Search Book
- Update Book
- Delete Book

Member Management

- Add Member
- View Members
- Search Member

Transaction

- Issue Book
- Return Book
- Fine Calculation

Reports

- Issued Books
- Available Books
- Member Report

8. Project Structure

```
Library_Management_System/
│── main.py
│── menu.py
│── login.py
│── books.py
│── members.py
│── transactions.py
│── reports.py
│── file_handler.py
└── utils.py
```

9. Features

- Book Management
- Member Management
- Issue & Return Books
- Fine Calculation
- Search Records
- Report Generation
- Role-Based Access
- File Handling

10. Future Enhancements

- SQLite/MySQL Database
- Barcode Scanner
- Flask Web Application
- React Frontend
- Email Notifications
- QR Code Integration
- Cloud Deployment

11. Conclusion

The Library Management System provides a structured and efficient way to manage library operations using Python. It demonstrates OOP, modular programming, functions, and file handling while serving as a foundation for future full-stack development.